Recall:

Meas: $(X, \Sigma_X \subseteq P(X))$ $f: X \to Y$ meas. if $\forall B \in \Sigma_Y$
$f^{-1}(B) \in \Sigma_X$

Meas. primitive, RVs derived

QBS: $(X, M_X \subseteq \mathbb{R} \to X)$ $f: X \to Y$ meas. if
$\forall \alpha \in M_X .\ f\alpha \in M_Y$

RVs primitive, measurability derived of subsets.

Meas $\hookleftarrow$ QBS

Main benefit QBS cartesian closed !

So QBS(X, Y) is itself a QBS, not just set.

Model of simply typed λ-calc. $(X \to Y) \to (Y \to X)$ makes sense

Today: Extend to program with distributions.

Idea: Use Giry monad:

GX = set of all prob. dist on X.

It self a QBS.

so eg. types like

$(X \to GY) \to G(Z \to GW)$

Recall semantics of 1st order language

Types: $A, B, A \times B, A + B \ldots$ were meas. space

Det. terms: $\Gamma \vdash^{d} t : A$ meas. maps

Prob terms: $\Gamma \vdash^{p} t : A$ were Markov kernels:

$[\![t]\!] : [\![\Gamma]\!] \to \Sigma_{[\![A]\!]} \to \mathbb{R}$ s.t

$\forall \gamma \in [\![\Gamma]\!]$ $[\![t]\!]_\gamma$ prob. meas. on $[\![A]\!]$

$\forall U \in \Sigma_{[\![A]\!]}$. $[\![t]\!]_{(-)}(U) : [\![\Gamma]\!] \to \mathbb{R}$ measurable.

Giry monad on Meas

Given (X, Σ_X) in Meas, let GX be set of all prob. meas. on X. Define Σ-alg. on GX as least such that $\forall A : \Sigma_X$. $ev_A : GX \to \mathbb{R}$ measurable.

$ev_A(\mu) = \mu(A)$

Thm: $\mathrm{Meas}[X, GY]$ corr. to Markov kernels.
Given $f : X \to GY$ meas $\bar{f} : X \to \Sigma_Y \to \mathbb{R}$ is
$\bar{f}\, x\, A = f\, x\, A$ also other way.

Thm: G is a monad:

$\delta : X \to GX \qquad \delta(x)[A] = \begin{cases} 1 & x \in A \\ 0 & \text{else} \end{cases}$

$+ \; f : X \to GY \mapsto \bar{f} : GX \to GY \qquad \bar{f}\mu = \mu \gg f$

$$\bar{f}\,\mu\,U = \int_{x \in X} f(x)\,U\,d\mu$$

Check: δ meas. & if f meas also $\bar{f}$ meas

+ axioms

Giry monad on QBS:

Definition 7: Recall, a prob. dist. on (X, M_X) is a pair (α, μ) where $\alpha : \mathbb{R} \to X$ is in M_X, μ: prob dist. on $\mathbb{R}$. Pushforward:

$\alpha_* \mu(A) = \mu[\alpha^{-1}(A)]$ for $A \in \Sigma_{M_X}$.

Integration: Given $f : (X, M_X) \to (\mathbb{R}, M_{\overline{\Sigma}_{\mathbb{R}}})$

$$\int f \, d(\alpha, \mu) = \int (f \circ \alpha)\, d\mu$$

Say $(\alpha, \mu) \sim (\alpha', \mu')$ if $\int (-)\, d(\alpha, \mu) = \int (-)\, d(\alpha', \mu')$

$$P(X, M_X) = \{(\alpha, \mu) \mid \alpha \in M_X, \mu \in G\mathbb{R}\} / \sim$$

$\alpha \in M_{PX}$ if $\exists \rho \in M_X$, $\exists g \in Meas(\mathbb{R}, G\mathbb{R})$

$\forall r \in \mathbb{R}.\ \alpha(r) = [(\rho, g(r))]$

Def 2: Let $\mathbb{W} = [0, \infty]$, $X^f = X \to 1$. Then

$$\int : \underbrace{(\mathbb{R} \to X^f)}_{RX} \to \underbrace{((X \to \mathbb{W}) \to \mathbb{W})}_{KX}$$

$$f \qquad g : X \to \mathbb{W} \mapsto \int_{x \in dom(f)} (g \circ f)\, dx$$

Let $DX = Im(\int) \subseteq KX$

with $\alpha : \mathbb{R} \to DX$ if $\exists \rho : \mathbb{R} \to RX$ meas. s.t.

$$\begin{array}{ccc} \mathbb{R} & \xrightarrow{\alpha} & DX \\ {\scriptstyle r}\downarrow & & \downarrow{\scriptstyle \int} \\ RX & \xrightarrow[\int]{} & KX \end{array}$$

So smallest QBS struct s.t. $\int$ is meas.

$PX = \{ \mu \in DX \mid \mu(\lambda x.\, 1) = 1 \}$